HW 2 鄭允翔 113705015

C105 01 Q13

13. • The data points (red dots) is distributed symmetrically to the blue line.

∴ The data is from normal population

C105 02 Q20

20. $H_0 = \mu = 74$

$H_\alpha = \mu \neq 74$

t-test

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{69.7 - 74}{16/\sqrt{25}} = -1.34 > -t_{24, 0.025} = -2.064$$

∴ Non-reject H_0, we don't have enough evidence to proof the significant difference of Average CBF in healthy people and stroke patients

C105 02 Q30

30. a. $H_0: \mu = 25$, $H_\alpha: \mu < 25$

$$t = \frac{20.3 - 25}{.5/\sqrt{21}} = -4.307 < -t_{20, 0.05} = -1.725$$

∴ Reject H_0, This is enough evidence to conclude that the true pretreatment mean is less than 25.

b. $26.6 \pm t_{20, 0.025} \times \frac{s}{\sqrt{n}} = 26.6 \pm 2.086 \times \frac{7.4}{\sqrt{21}}$

$= 26.6 \pm 3.368 \Rightarrow (23.23, 29.97)$

c. The self-esteem score increased from pretreatment to post-treatment and follow ups. This means that after going through the treatment, the self-esteem increased which means the treatment is successful.

JOYKO® 36 Lines, 6 mm

C10S02Q35

35. a. $H_0: \mu = 22$, $H_\alpha: \mu > 22$

$$t = \frac{31 - 22}{6.2/\sqrt{17}} = 5.985 > t_{16, 0.05} = 1.746$$

∴ Reject H_0, there's enough evidence to proof that injected rats drink more water than uninjected rats

b. $31 \pm \underbrace{t_{16, 0.05}}_{1.746} \times \frac{6.2}{\sqrt{17}} = 31 \pm 2.625 \Rightarrow (28.37, 33.63)$

C10S03Q7

7. $\bar{x}_1 = \frac{12+3+8+5}{4} = 7$

$$S_1^2 = \frac{(12-7)^2 + (3-7)^2 + (8-7)^2 + (5-7)^2}{3} = 15.33$$

$\bar{x}_2 = \frac{14+7+7+9+6}{5} = 8.6$

$$S_2^2 = \frac{(14-8.6)^2 + (7-8.6)^2 + (7-8.6)^2 + (9-8.6)^2 + (6-8.6)^2}{4} = 10.3$$

$H_0: \mu_1 - \mu_2 = 0$, $H_\alpha: \mu_1 - \mu_2 \neq 0$

$df = n_1 + n_2 - 2 = 4 + 5 - 2 = 7$

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{df} = \frac{3 \times 15.33 + 4 \times 10.3}{7} = 12.457$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{S_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{7 - 8.6}{\sqrt{12.457\left(\frac{1}{4} + \frac{1}{5}\right)}} = -0.675 > -t_{7, 0.025} = -2.365$$

R.R : $|t| > t_{7, 0.025} = 2.365$

p-value : $2 \times P[t < -0.675] \approx 0.52 > 0.05$

∴ Non-reject H_0, our t-score lies between the boundaries of -2.365 to 2.365 and our p-value is way higher than 0.05.

C10S03Q8

8. $H_0: \mu_1 - \mu_2 = 0$ $H_\alpha: \mu_1 - \mu_2 \neq 0$

$df = 16 + 13 - 2 = 27$

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{df} = \frac{15 \times 4.8 + 12 \times 5.9}{27} = 5.288$$

No
Date

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} = \frac{34.6 - 52.2}{\sqrt{5.288\left(\frac{1}{16}+\frac{1}{13}\right)}} = 2.795 > t_{27,\,0.025} = 2.052$$

R.R: $|t| > t_{27,\,0.025} = 2.052$

P-value $= 2 \times P(t > 2.795) \approx 0.0094 < 0.05$

$\therefore$ Reject H_0, t-score is outside of the boundaries and p-value is lower than 0.05

CIDSO3 Q12

12. a. $H_0: \mu_1 - \mu_2 = 0$, $H_A: \mu_1 - \mu_2 \neq 0$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}} = \frac{86.4 - 79.7}{\sqrt{\frac{5.95^2}{10}+\frac{10.7^2}{10}}} \approx 1.5188 < t_{18,\,0.025} = 2.101$$

$\therefore$ Non-reject H_0, t-score is in range, so ~~[illegible]~~ there's no significance difference

b. $86.4 - 79.7 \pm \underbrace{t_{18,\,0.025}}_{2.101} \times \sqrt{\frac{5.95^2}{10}+\frac{10.7^2}{10}} \Rightarrow (-2.568,\ 15.968)$

c. From part a's conclusion and 95% CI, we can't tell if IPad is better or worse than traditional book because there's no significance difference